

C.V RAMAN GLOBAL UNIVERSITY

MID-SEMESTER EXAMINATION 2025-26(1st)

Prog: B.Tech
Sub. Code: PH101

Subject :-PHYSICS
Time: 1 Hour

Semester:1st GROUP-1,3,5,7,9

Branch: ALL
Max marks: 20

The figures in the right-hand margin indicate marks.

PART A

Q1. Answer ALL questions:

(5x2)

- a The differential equation of motion of a freely oscillating body is given by

CO1

BL-2

$$2\frac{d^2x}{dt^2} + 18\pi^2x = 0$$

Calculate the time period and frequency of the oscillation.

- b In a plane diffraction grating of there are 5000 lines in 1 mm width. Find the grating element of the diffraction grating.

CO1

BL-3

- c A glass plate having refractive index 1.66 is to be used as a polarizer. Calculate polarizing angle and angle of refraction.

CO1

BL-3

- d Find the divergence of curl of the following vector.

CO2

BL-3

$$\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}.$$

- e Write the relation between displacement vector and electric field. What is the unit of displacement vector?

CO2

BL-2

(2 x5)

PART-B(Answer as per instructed)

- Q2 Derive the differential equation for a damped harmonic oscillator. Write the conditions for different cases of damped oscillation.

CO1

BL-3

OR

With a suitable labelled diagram explain the formation of Newton's ring as interference of light. Derive an expression for the diameter of bright rings in Newton's ring experiment.

Q3

Show that (i) $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ (ii) $\vec{\nabla} \times \vec{B} = \mu_0 \left[\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right]$

CO2

BL-3

OR

Write the integral forms of Maxwell's electromagnetic equations in dielectric media. Write the significance of Maxwell's equations.



Registration No. :

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Total number of printed pages – 03

B.Tech

Subject Code: PH101

1st SEMESTER REGULAR EXAMINATION – 2025-2026

SUBJECT NAME: PHYSICS

BRANCH: Gr-1, Gr-3, Gr-5, Gr-7 and Gr-9

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	A particle performs SHM described by $x(t)=0.12 \cos (4t)$. What is the displacement at $t= 1$ sec.	1	2	1
(b)	What do you mean by path difference and phase difference in interference? How are they related?	1	2	3
(c)	Write any two important differences between the e-ray and the o-ray.	1	2	2
(d)	State Gauss's law for magnetic fields. Write its integral forms.	2	2	1
(e)	Write the condition for a vector to be irrotational.	2	2	2
(f)	Calculate the volume charge density at $(1,1,0)$ due to electric field $\vec{E} = 2xy \hat{i} + y^2 \hat{j} + 3xz \hat{k}$ in vacuum ($\epsilon_0 =$ Permittivity of vacuum).	2	2	3
(g)	Write the expression of momentum and energy operators.	3	2	1
(h)	Find the eigen value for the operator $\frac{d^2}{dx^2}$ in the eigen function $\Psi(x) = 2e^{2x}$.	3	2	2
(i)	A ball of mass 1 gm has a speed of 100 m/s. Calculate the de-Broglie wavelength associated with it.	3	2	3
(j)	Plot graphs of the Fermi-Dirac probability distribution function versus energy at temperatures $T=0$ and $T>0$	4	2	2
	Why the term 'bipolar' is used in bipolar junction transistor?	4	2	1

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|-----|---|---|---|---|
| (l) | Calculate the Hall voltage when the magnetic field is 8 A/m, current is 4 A, width is 0.005 m and the concentration of carrier is $10^{22}/m^3$. | 4 | 2 | 3 |
| (m) | How LASER light is different from ordinary light. | 5 | 2 | 2 |
| (n) | What do you mean by cooper pair? | 5 | 2 | 1 |
| (o) | Define numerical aperture of an optical fiber. Calculate the numerical aperture of an optical fiber whose core and cladding are made of materials of refractive index 1.6 and 1.5 respectively. | 5 | 2 | 3 |

Part-B

C
O Marks

Q2. Answer any two:

- | | | | |
|-----|--|---|-----|
| (a) | Starting from the differential equation for a damped harmonic oscillator discuss underdamped, critically damped, overdamped oscillations. Represent these oscillations on displacement time graph. | 1 | 7 |
| (b) | With a suitable labelled diagram explain the formation of Newton's ring as interference of light. Derive an expression for the diameter of dark rings in Newton's ring experiment. | 1 | 4+3 |
| (c) | State Malus' law. Find the angle between the axis of an analyser with incident beam so that the intensity of the polarized light is 25% less than that of initial intensity. | 1 | 2+5 |

Q3. Answer any two:

- | | | | |
|-----|--|---|-----|
| (a) | Use Gauss's law to obtain the expression for the electric field due to a point charge and hence derive Coulomb's law. An electron and a proton is placed inside a sphere of radius 15cm. Calculate the net flux coming out of the surface of the sphere. | 2 | 4+2 |
| (b) | State Faraday's laws of electromagnetic induction? Derive $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ using Faraday's laws of electromagnetic induction | 2 | 3+ |
| (c) | Define and derive displacement current density in terms electric field intensity. In free space electric field intensity is given as $\vec{E} = \hat{y} 20 \cos(\omega t - 50x)$ volt/meter. Calculate the displacement current density. | 2 | 3+ |

Q4. Answer any two:

- | | | | |
|-----|---|---|---|
| (a) | Define de-Broglie hypothesis for matter wave. Derive the relation between de Broglie wavelength and accelerating potential V for an electron. The de Broglie wavelength of a particle moving with a velocity of 200 m/s is 14.14 Å. What will be the de Broglie wavelength if the kinetic energy is four times? | 3 | 2 |
| (b) | Write the Schrödinger equation for a particle inside an infinite potential well and derive energy eigen values and eigen functions. | 3 | |
| (c) | What is expectation value? Write the expression for expectation value of momentum. The probabilities that a system can be in the states represented by eigen functions ψ_1, ψ_2, ψ_3 are 1/2, 1/3, 1/6 respectively. If | 3 | |

the energy eigen values for the above states are 4 eV, 6 eV and 9 eV, respectively, find the energy expectation value.

Q5. Answer the followings:

- (a) Derive the expression for the carrier concentration of holes in valence band in intrinsic semiconductor. 4 7
- (b) Discuss working principle of PN junction diode with a schematic diagram. Plot IV characteristics of a PN junction diode in both forward and reverse bias mode. 4 5+2

Q6. Answer the followings:

- (a) Write the properties of Laser light. Explain basic working principles of typical Laser source of light. Give four important uses of Laser. 5 4+1 1
- (b) Explain Meissner's effect with suitable diagram? What is Type I and Type-II superconductor. 5 4+3 2

Best of Luck

$$D_{12} = \frac{2\pi}{\hbar} \frac{1}{\omega} \cos \theta$$

$$\frac{2\pi}{\hbar} \times \frac{1}{\omega} = \frac{2\pi}{\hbar} \cos \theta$$

$$\cos \theta = \sqrt{\frac{1}{4}}$$

$$\theta = \sqrt{\frac{1}{4}}$$

$$J = \frac{D}{S}$$

$$D = \frac{q}{S}$$

$$= \frac{\int \rho dw}{S}$$

$$D = J_D = \frac{d\sigma}{dt}$$

$$= \epsilon_0 \frac{dE}{dt}$$